A_GMX on crowded CCDs

background override, serial-register streaks, straight-track hints, crossing split and track linking

comm_scripts clustering study

code of 29 October 2026 (`clustering_tests/crowded.py`)

Abstract

On CCDs where tracks cover a large part of the image (surface runs such as L2 ACM101, the Am-241 test CCD), A_GMX had three failures: its row-by-row background fit picked the wrong whole-electron pedestal in rows with $\sim 10\,e^-$/pix of dark charge; serial-register streaks were merged into tracks; and long straight tracks (muons) were cut into several clusters wherever they crossed other tracks, while touching tracks were merged into star-shaped clusters of up to 10^4 px. *Crowded mode* fixes these and is switched on per image when the image is crowded. The straight tracks are found by a separate finder, but only their *axis* is passed to A_GMX as a hint: the cluster's width, footprint and the charge shared with crossing tracks remain A_GMX's own. On L2, injected muons go from 4.1 fragments and 45 % of their charge in one cluster to 1.9 fragments and 93 %; Fe-55 Kα narrows from 220 to 85 eV on the worst CCD and gains 14–65 % events. On the Am-241 crowded CCD (EXT4) the energy scale of the constrained 10–30 keV fit is unchanged ($k = 1.0004$ vs 1.0009 on the same images), and on EXT3 too ($k = 0.9952 \pm 0.0008$ vs 0.9964 ± 0.0010, 29 images) once the fit is started from narrow line widths (section 3.5). Crowded mode is the default (`crowded='auto'`): it runs only on images that pass the trigger, and other images are clustered as before.

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1 When is an image crowded?

The trigger is the *core fill*: the fraction of active pixels whose charge, smoothed with a Gaussian of 1 px, exceeds $\max(60\,e^-,\, 5\,\sigma_\lambda)$, where $\sigma_\lambda = \sqrt{\lambda}/(2\sqrt{\pi})$ is the noise of the smoothed dark charge λ. It measures how much of the CCD is covered by deposits, independent of the background level.

image type	core fill	crowded mode with `auto`
L2 ACM101 surface run (EXT1/2/4)	0.27–0.31	on
Am-241 AMTest EXT3 / EXT4	0.27 / 0.43–0.67	on
Fe-55 production CCDs (gn_lowcharge_study)	0.015–0.02	off

`crowded='auto'` switches crowded mode on for core fill ≥ 0.10; `'on'` and `'off'` force it. Below the threshold nothing in this note runs, and A_GMX output is identical to the code without crowded mode (unit test).

2 The five pieces

2.1 Background override (only at high dark charge)

A_GMX's background model fits each row with a Poisson⊗Gaussian (offset o, dark charge λ, noise s). With $\lambda \gtrsim 5\,e^-$/pix the zero-electron peak is gone and o is only known modulo whole electrons: on L2 EXT1, 395 of 500 rows took a pedestal 4–85 e^- too high, which made energies wrong by up to $-6\,e^-$/pix × the cluster area and produced a 10–160k px "blob". The override:

1. is applied only for electron-counting images (median $s < 0.45\,e^-$) with median $\lambda \geq 3\,e^-$/pix;
2. flags a row as bad when more than 0.5 % of its pixels lie below $o - 0.5 - 3s$, and gives it the whole-electron pedestal of the good rows (keeping its fractional part);
3. re-estimates λ in cells of 10×64 px by a truncated-Poisson fit below the 90 % quantile (so that the halos of tracks do not raise it), and overrides the row model where the two differ by more than $\max(0.3,\, 5\sigma)\,e^-$/pix; bad rows take the cell value.

The gate on λ matters: on Am-241 ($\lambda = 0.2$–$2.4\,e^-$/pix) the dense X-ray field fools both tests (EXT3: 40–70 % of the cells changed, energies −0.2 %, 59.5-keV events lost; EXT4: 429 "bad" rows and backgrounds of $10^4\,e^-$/pix). A unit test checks that the override never moves a row's pedestal away from the truth.

2.2 Serial-register streaks

Charge clocked into the serial register lands in one row with no vertical spread. A significant pixel is *single-row* when the pixels above and below hold < 25 % of its charge; horizontal runs of ≥ 8 such pixels (1-px gaps bridged; pieces ≤ 10 px apart in a row joined) are removed from the image before clustering and reported as clusters with class `artifact-row` and regime `serial` (`agmx_regime` = 3).

2.3 Straight-track hints

Finding. The *core* (as in section 1) is reduced to its 1-px skeleton; a Hough transform in tiles of 160×160 px (half overlapping; votes summed over 3 neighbouring ρ bins so that a wobbling skeleton

still counts) gives candidate lines with ≥ 40 votes. Lines within 10° of a row are left to the streak finder. Each candidate is refined (charge-weighted PCA of the core pixels within 2 px) and walked along the image in both directions. A step is occupied when a core pixel within ± 1 px of the axis is a transverse ridge (its smoothed charge at least twice the smaller of the two values at ± 3.5 px) or lies on a track already accepted (so crossings do not stop the walk); the walk ends after 6 empty steps.

Accepting. A line is a straight track if it is ≥ 40 px long, $\geq 85\,\%$ occupied, its median transverse rms is ≤ 2.2 px, it is not a duplicate, and its charge is *continuous*: of the steps along the axis, at most 15 % may lie below half and at most 8 % below 0.3 of the median axis charge. The continuity test separates muons (injected: 90th percentile 0.09) from chains of X-ray blobs in a dense field (Am-241: 10th percentile 0.17); without it the finder took 166 "tracks" on an Am-241 image, with it 13 (figure 7).

Using the hints. Nothing is removed from the image. A_GMX runs as always (seeds, EM of the Gaussian clouds, merge graph, island arbitration). Then, for each hint (axis centre $\mathbf{c}$, direction $\mathbf{u}$, extent $t_0..t_1$, rms width w), every cloud k with

$$|(\boldsymbol{\mu}_k - \mathbf{c}) \cdot \mathbf{n}| \leq \max(1.5,\, w), \qquad t_0 - 3 \leq (\boldsymbol{\mu}_k - \mathbf{c}) \cdot \mathbf{u} \leq t_1 + 3$$

($\mathbf{n} \perp \mathbf{u}$) goes into one cluster, unless it is elongated (axis ratio ≥ 1.6) at more than 30° to the hint, which makes it part of a crossing track. The cluster gets class `muon` and `straight_track = True`. Because the pixels' charge is still split by the mixture's responsibilities, a crossing track keeps its charge and stays one cluster, and the muon's width is the width A_GMX fits (figure 3).

An earlier version cut a tube around each track out of the image and shared charge at crossings with an EM step of its own; its tubes were twice as wide as the tracks (its width estimate collected neighbours and halos), it took X-rays lying on the muon's path (5–8 % of the Fe-55 events on L2) and left crumbs. The hint version has none of these.

2.4 Crossing split

After the merge graph and after the island arbitration, a cluster whose clouds hold two straight lines (RANSAC on the cloud centres, each ≥ 20 px and ≥ 3 clouds) that cross as an X ($\geq 20^\circ$, both continuing ≥ 8 px past the crossing) is split into the two tracks; clouds go to the nearer line.

2.5 Track linking

Collinear straight fragments of one track are joined (both pieces ≥ 12 px, endpoints within $\max(2.5,\, 1.5 + 0.01L)$ px of the longer piece's axis, parallel within 4°, gap < 15 px, or < 25 px for pieces ≥ 30 px). Shorter pieces — photo-electron tracks of 59.5-keV gammas, X-rays — are never joined (with 5 px, Am-241 EXT3 lost 4 % of its 59.5-keV events). Streaks are not linked.

3 Results

3.1 L2 surface run: maps

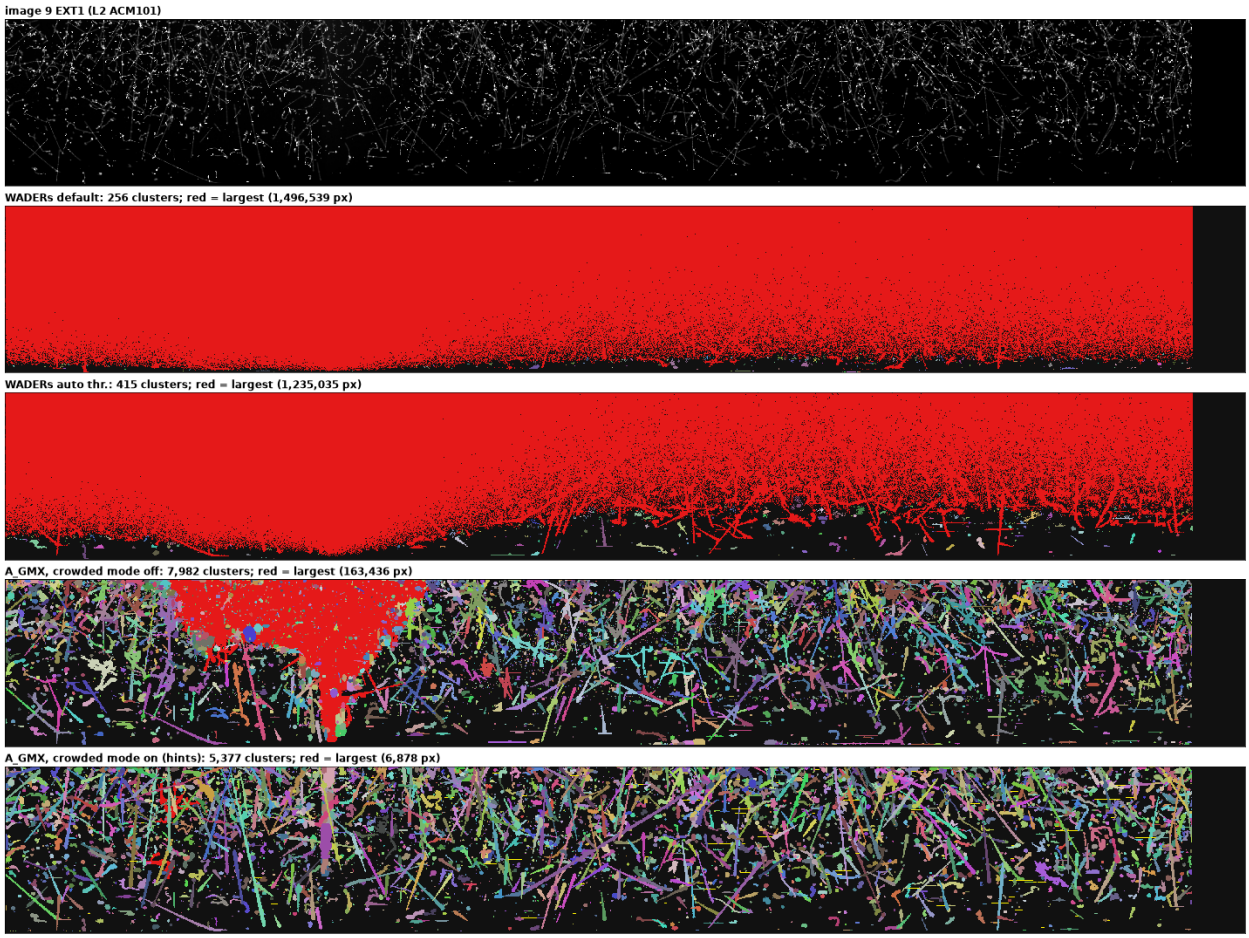


Figure 1: L2 image 9 EXT1. From the top: image; WADERs (default and electron thresholds: one cluster covers most of the CCD); A_GMX without and with crowded mode. Red: the largest cluster; yellow: serial-register streaks.

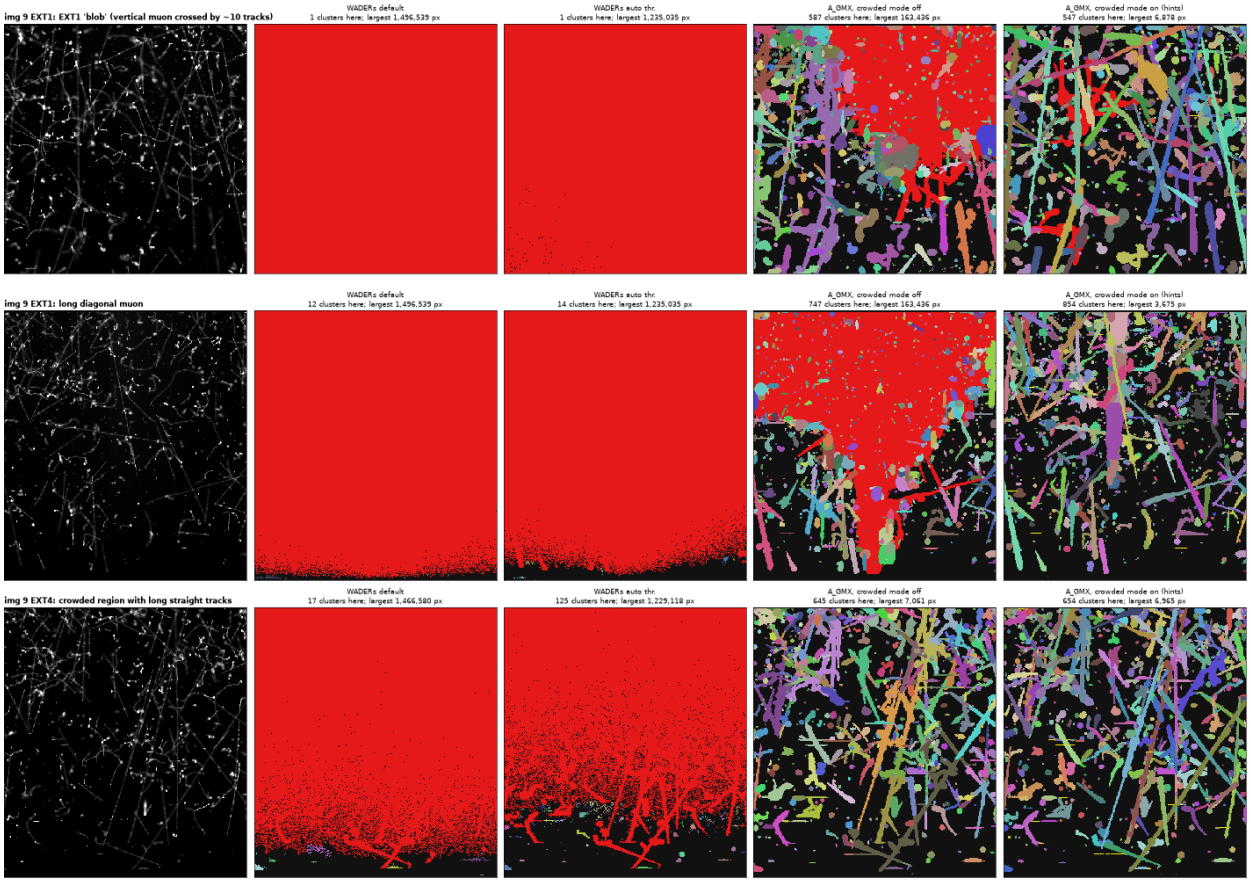


Figure 2: Close-ups: the EXT1 "blob" (a vertical muon crossed by about ten tracks), a long diagonal muon, a crowded region of EXT4. Columns as above; titles give the number of clusters in the cut-out and the largest.

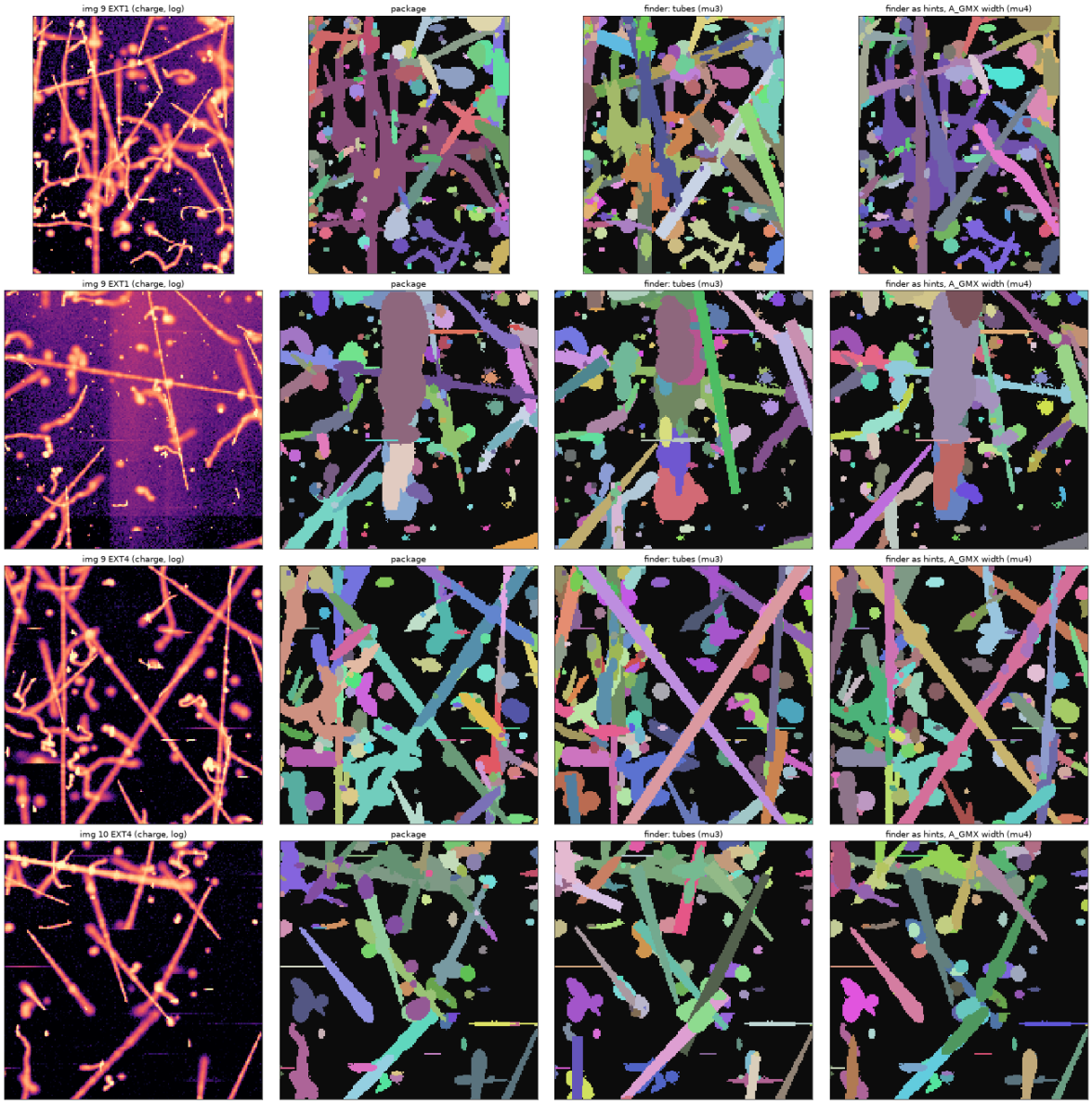


Figure 3: Track width. Columns: charge; the package without straight tracks; the earlier tube version (too wide); the hint version, where the width is A_GMX's own.

3.2 L2: injected muons

25 muons per image (projected length 40–400 px, Landau-fluctuating charge, diffusion growing from 0.2 to 0.8–1.6 px with depth) were added to the real L2 images 9 and 10 (EXT1, EXT4; 2 seeds: 200 muons) and the images clustered.

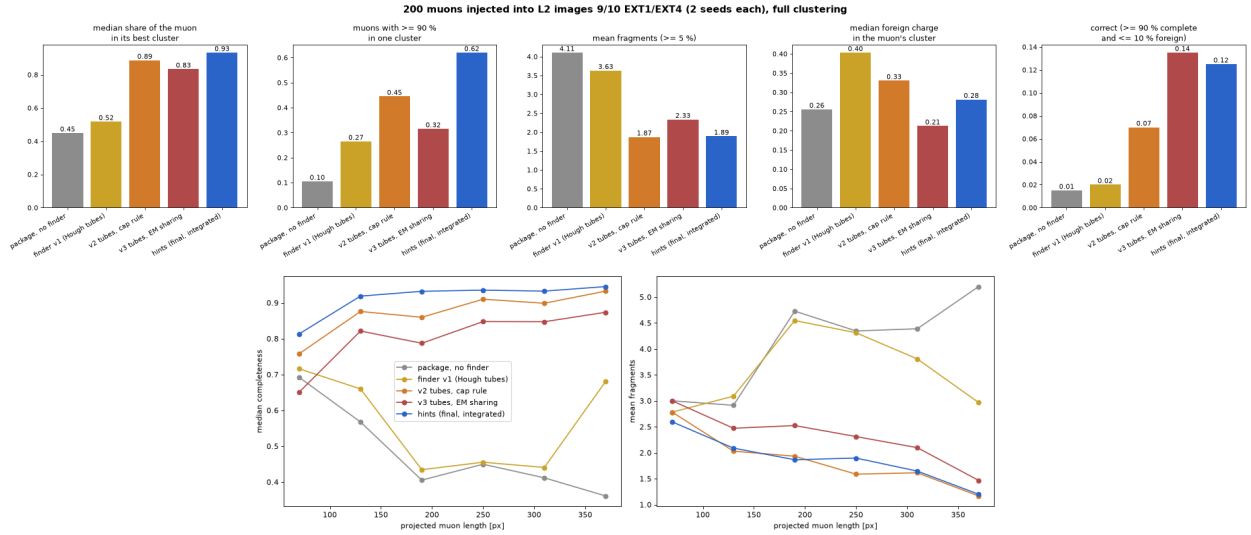


Figure 4: Per muon: share of its charge in its best cluster, fraction with $\geq 90\,\%$ in one cluster, number of fragments ($\geq 5\,\%$), foreign charge in its cluster, and "correct" ($\geq 90\,\%$ in one cluster and $\leq 10\,\%$ foreign). The last bar is the final, integrated code. In this image a 300-px line crosses about 15 other tracks, so few muons can be "correct".

	no straight tracks	tubes (earlier)	hints (final)
median share in the best cluster	0.45	0.83	0.93
$\geq 90\,\%$ in one cluster	10 %	32 %	62 %
fragments per muon	4.1	2.3	1.9

3.3 L2: Fe-55

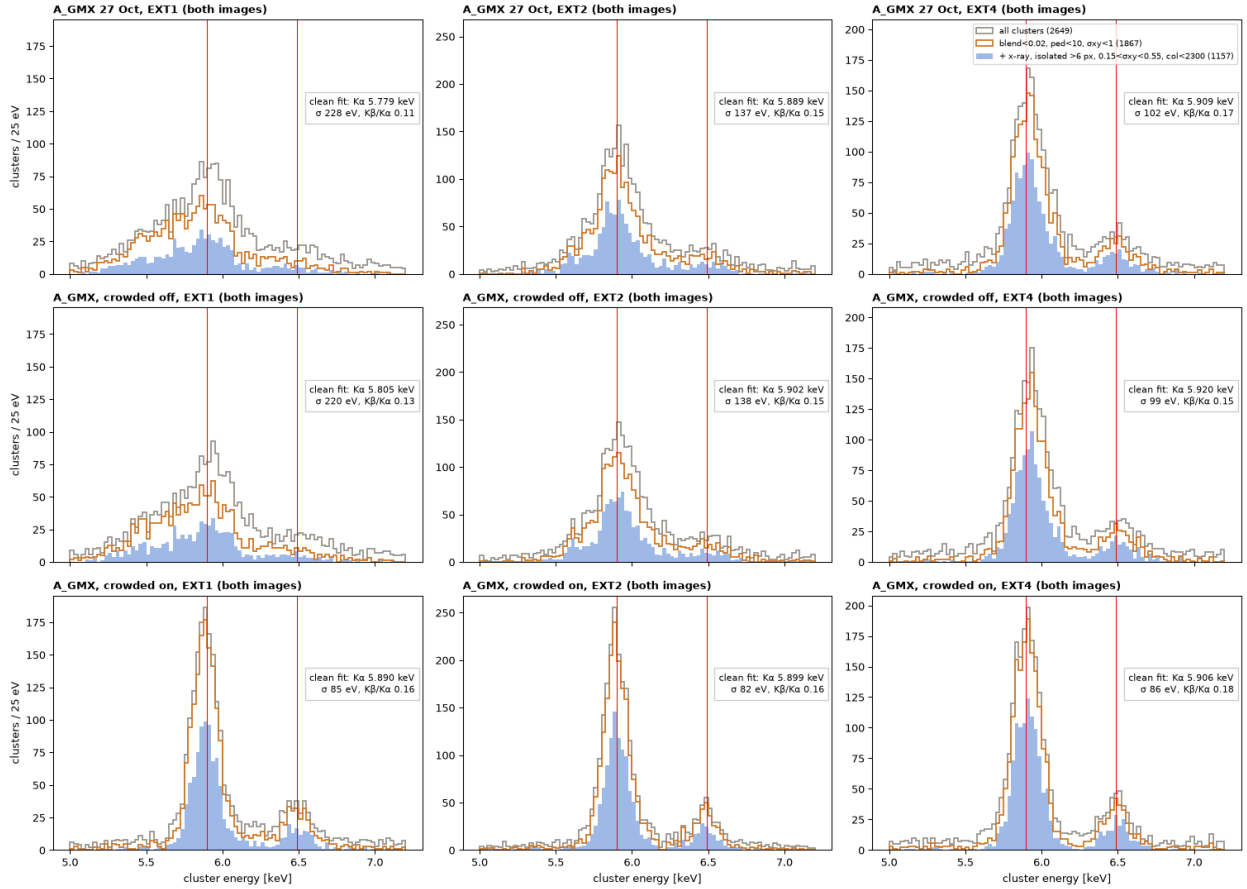


Figure 5: Fe-55 on L2 (images 9 and 10), front side. Rows: A_GMX of 27 October; today's A_GMX with crowded mode off; with crowded mode on. Blue: clean selection (blend < 0.02, ped < 10, $\sigma_{xy} < 1$, x-ray class, isolated > 6 px, $0.15 < \sigma_{xy} < 0.55$, column < 2300). Energies from 29 October use $\varepsilon(130\,\mathrm{K}) = 3.7486\,\mathrm{eV}$.

clean Kα events / width	EXT1	EXT2	EXT4
crowded off	476 / 220 eV	797 / 138 eV	896 / 99 eV
crowded on	785 / 85 eV	1016 / 82 eV	1021 / 86 eV

3.4 Am-241, EXT4 (the crowded CCD)

Ten of the 29 AMTest images (every third) were reclustered with crowded mode on and compared with the A_GMX clusters of 27 October on the same images (energies scaled to the same ε).

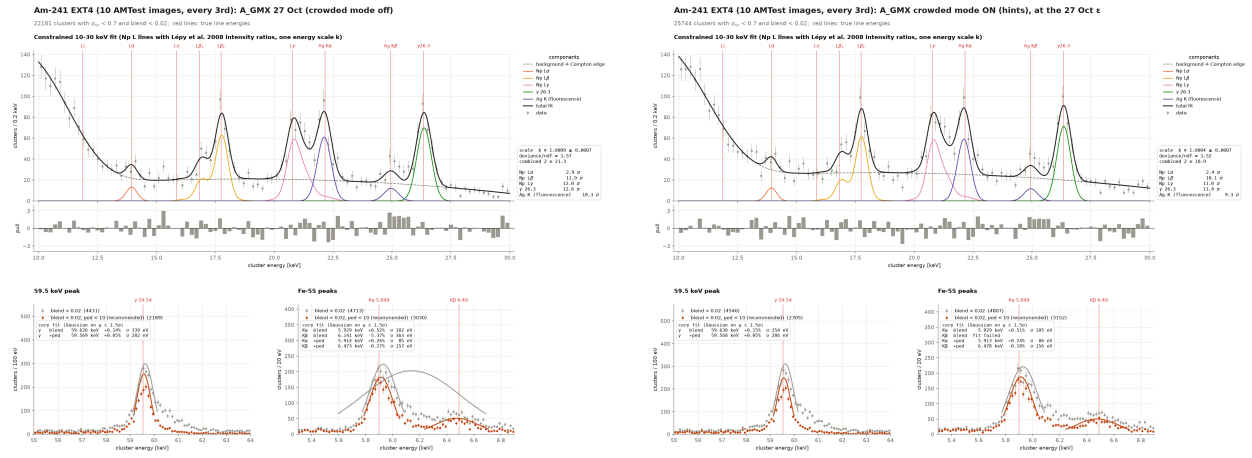


Figure 6: Constrained 10–30 keV fit (Np L lines with Lépy et al. intensity ratios, one energy scale k), 59.5-keV and Fe-55 peaks. Left: without crowded mode ($k = 1.0009 \pm 0.0007$, $Z = 21.2$, 22 181 clusters); right: with crowded mode ($k = 1.0004 \pm 0.0007$, $Z = 19.3$, 25 744 clusters).

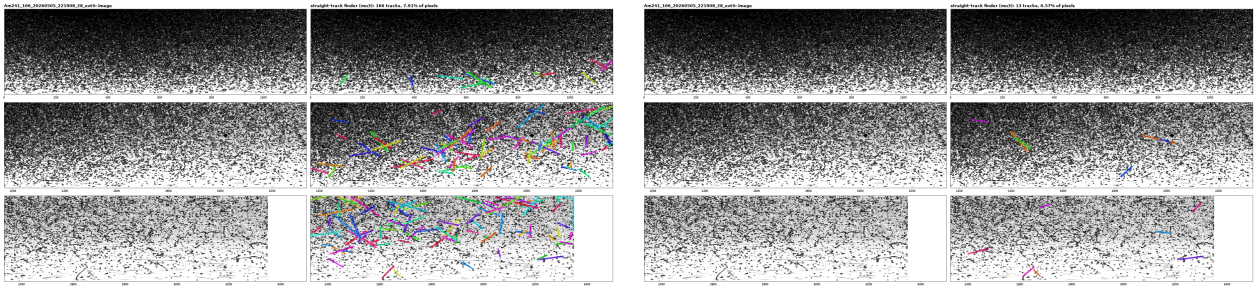


Figure 7: Straight tracks on an Am-241 EXT4 image without (left, 166) and with (right, 13) the continuity test.

3.5 Am-241, EXT3

With crowded mode the fit routine (`amfit.Model.fit`, 9 starts) first returned broad lines at the upper bound of the width ($s_0 = 0.8$ keV, $k = 0.973$, $Z = 4.7$). That was a local minimum: starting from narrow widths held fixed and then freed gives a likelihood higher by $\Delta \ln L = 42$ with $s_0 = 0.05$ keV, the same as without crowded mode. The figures use this extended start. Crowded mode adds 7–12 % events to the Np L lines and 26.3 keV and removes 2.8 % at 59.5 keV (with the 59.5-keV width 21 eV narrower).

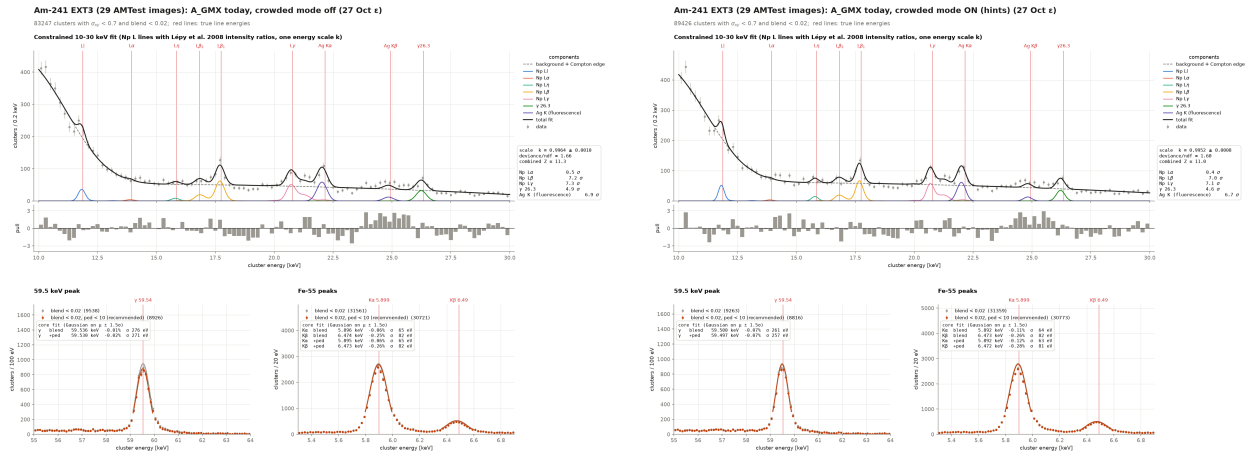


Figure 8: EXT3, 29 images, today's code with crowded mode off (left: $k = 0.9964 \pm 0.0010$, deviance/ndf 1.66, $Z = 11.3$) and on (right: $k = 0.9952 \pm 0.0008$, 1.60, $Z = 11.0$). The 59.5-keV and Fe-55 peaks move by 3–33 eV.

3.6 Energy of the straight tracks

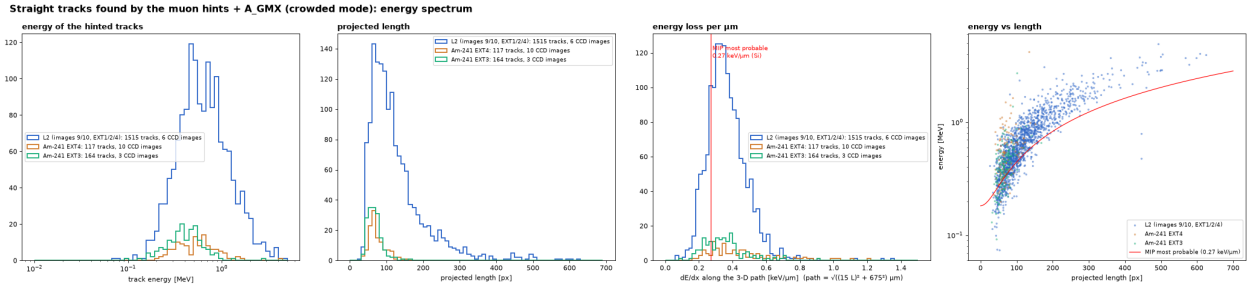


Figure 9: Clusters defined by the hints: energy, projected length, energy loss along the 3-D path $\sqrt{(15L)^2 + 675^2}$ µm and energy vs length. On L2 (1515 tracks) the median energy is 0.61 MeV and the most probable energy loss 0.34 keV/µm, ∼25 % above the 0.27 keV/µm of a minimum-ionising particle: the clusters also carry their share of the tracks they cross, and some are straight stretches of electron tracks.

4 Open items

- **Am-241 EXT3:** 59.5 keV loses 2.8 % of its events with crowded mode (not traced to one piece).
- **Fit routine:** `amfit.Model.fit` can end in a wide-line minimum; it should also start from narrow widths.
- **Electron tracks:** straight 50–90 px stretches of curly electron tracks are taken as straight tracks.
- **Muon energy:** the hinted clusters' energy loss is 25 % above the MIP value on L2; how much of it is charge of crossing tracks has not been measured.
- **Validation:** the injected muons are simulated (Moyal fluctuations, linear diffusion); EXT4 was checked on 10 of 29 images.

5 How to run

```
run_WADERs images.txt --gain-correction --clusterer agmx --agmx-crowded auto # crowded
    mode where the core fill >= 0.10
run_AGMX WADERs_analysis_<run> --agmx-crowded on # force it on a finished run
AGMX_CROWDED=auto python my_script.py # in python: environment default
```

```
import adaptive as AD
clusters, info = AD.adaptive_cluster(Q, valid, engine="gmx", crowded="auto", return_info=
    True)
info["core_fill"], info["crowded_mode"], info["n_hints"], info["n_streaks"], info["
    n_linked"], info["bg_override"]
```

Each output file records the setting and what ran in `agmx_config` (key `crowded_mode`). Code: `clustering_tests/crowded.py` (the five pieces), `adaptive.py` (the switch), `gmx.py` (`general_cluster(split_crossings=, hints=)`); tests: `tests/test_agmx_crowded.py`; the study that led here: `wip/agmx_crowded/`.